

COMPSYS 305: Mini Project

Design Report

1. Game Strategy

Our project implements a real-time hardware "Flappy Bird" on the DE0-CV FPGA, focusing on hardware-level physics and VGA synchronization.

- **Vertical Physics:** Gravity is simulated by increasing the per-frame vertical velocity (v_y). When the PS/2 mouse clicks, an upward impulse is applied, resulting in a "flap" behavior.
- **Collision Detection:** Overlapping real-time (x, y) coordinate comparisons between birds and pipes/grounds trigger a GAME_OVER state.
- **Difficulty Scaling:** To meet project requirements, adjust the speed and pipe spacing height dynamically across three steps based on the current score.
- **LFSR Randomization:** Linear Feedback Shift Register (LFSR) provides pseudo-random pipe heights for different gameplay.
- **VGA Synchronization:** All logic updates occur during V_Sync to ensure flicker-free rendering.

2. Block Diagram

3. State Machine

Our team's game control logic is designed with a Moore-style Finite State Machine (FSM) for hardware efficiency. The entire system has four main states, each of which is switched by the user's input (button, mouse) and in-game signals (collision detection).

i. State Definitions

- a. IDLE (Initial Screen): The state of the system being powered on or immediately after reset.
 - Screen output: A standby message such as "Press Start" or "Hello World" is displayed on the screen.
 - Operation: The position of all obstacles (Pipes) and the coordinates of the bird are fixed to their initial values.
- b. PLAYING: The actual game is active.
 - Operation: Gravity logic is applied to lower the bird, and the vertical speed is updated each time a mouse-click signal is received. The pipe moves to the left, and the score counter is activated.
- c. PAUSED:
 - Description: Enter when the user presses a certain button (SW2) during the game.
 - Operation: All coordinate updates by frame_tick are interrupted, leaving the screen frozen.
- d. GAME_OVER:
 - Description: Enter when the bird hits the pipe or hits the ground (bird_dead = '1').
 - Operation: Display "GAME OVER" with large text (scaling applied) on the screen and secure the final score to the HEX display.

ii. State Transition Logic

- IDLE → PLAYING: Toggle when the user presses the Start button (SW1).
- PLAYING → PAUSED: Toggle when pressing the Pause button (SW2) during the game.
- PAUSED → PLAYING: Press the Pause button again to resume the game.
- PLAYING → GAME_OVER: Switching when the collision_event signal is generated from the collision detection module.
- GAME_OVER → IDLE: Return to the initial screen after a reset signal has been entered or a certain period of time.